DECOUPLING: place each 100n at its supply pin

TVP5150 IO_DVDD



TVP5150 DVDD



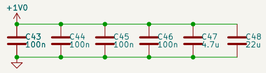
Flash



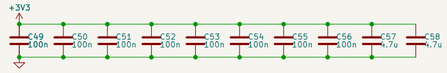
27 MHz osc



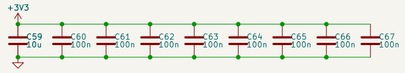
FPGA VCC (1V0)



FPGA VCCX / VCCIO (3V3)



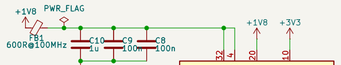
FT602 VCC33 / VDDA / VCCIO (3V3)



COMPOSITE VIDEO INPUT

75R termination split 37.4R + 37.4R gives a 2:1 pad (TVP5150 input max 0.75 Vpp).
TI recommends an anti-aliasing low-pass filter (~6 MHz) before the coupling cap: not fitted.

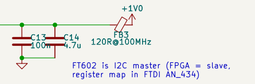
Analog 1V8 (CH_AVDD, PLL_AVDD)



VCCPLLL1 filter



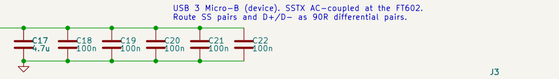
VCCPLLR1 filter



FT602 is I2C master (FPGA = slave, register map in FTDI AN_434)

DV10 = internal 1V0 LDO output -> VD10 / AVDD

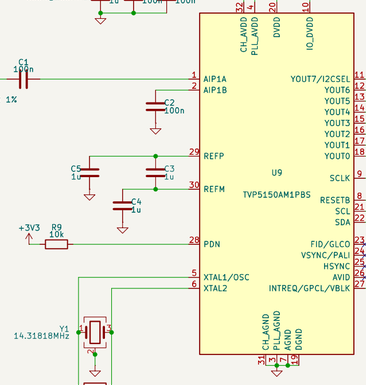
USB 3 Micro-B (device), SSTX AC-coupled at the FT602. Route SS pairs and D+/D- as 90R differential pairs.



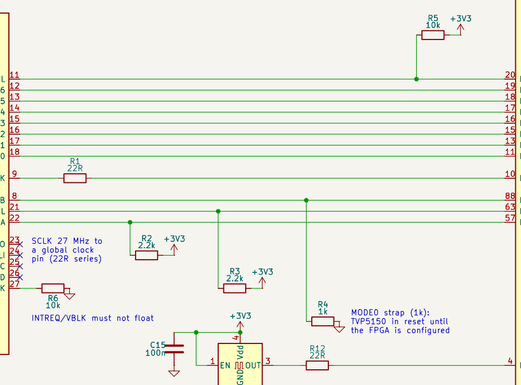
J1 RCA (composite)



U9 TVP5150AM1PBS



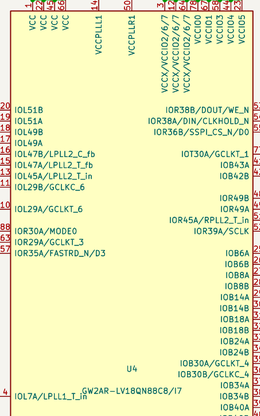
14.31818 MHz, CL 18 pF (27p each side)



SCLK 27 MHz to a global clock pin (22R series)

INTREQ/VBLK must not float

U4 GW2AR-LV18QN88C8/I7

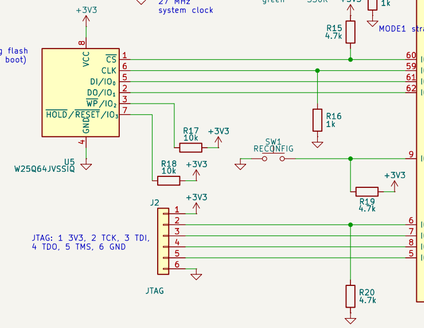


MODE0 strap (1k): TVP5150 in reset until the FPGA is configured

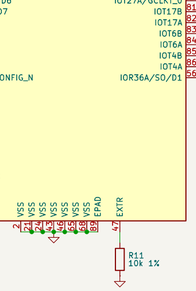
27 MHz system clock

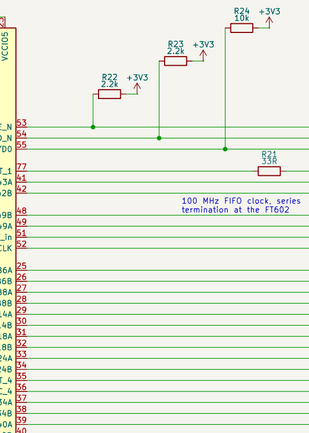


Config flash (MSPI boot)



JTAG: 1 3V3, 2 TCK, 3 TDI, 4 TDO, 5 TMS, 6 GND





R5 10k +3V3

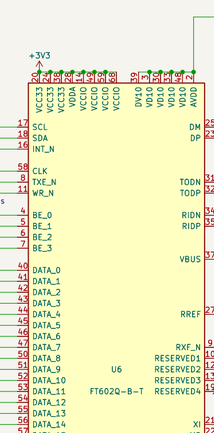
R22 2.2k +3V3

R23 2.2k +3V3

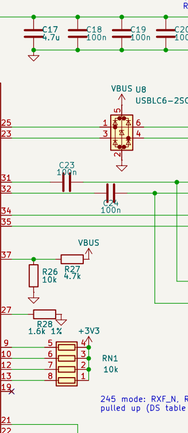
R24 10k +3V3

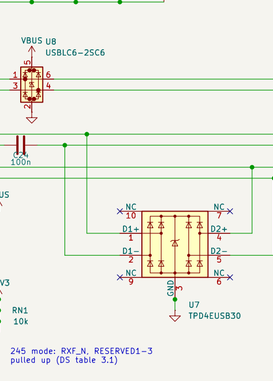
R21 33R

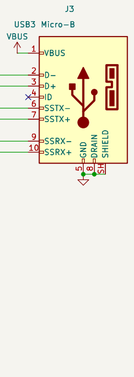
100 MHz FIFO clock, series termination at the FT602



U6 FT602Q-B-T







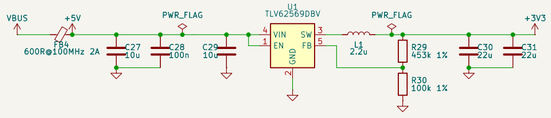
R25 10k +3V3

C16 100n

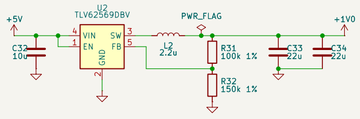
FT602 runs even with the FPGA blank

POWER: USB VBUS 5 V -> 3V3 buck (2 A), 1V0 buck (2 A, FPGA core), 1V8 LDO (TVP5150)

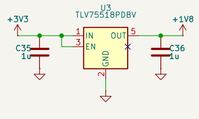
Budget: FT602 ~185 mA @ 3V3 (less at SD), TVP5150 ~60 mA @ 1V8, GW2AR-18 core 100–300 mA @ 1V0.
All rails come up together from 5 V, as on the Tang Nano 20K.



3V3: Vout = 0.6 V x (1 + 453k/100k)



1V0: Vout = 0.6 V x (1 + 100k/150k)



1V8 (TVP5150 DVDD, AVDD via ferrite)

5150UVC: RCA composite video -> TVP5150AM1 -> GW2AR-18 FPGA -> FT602 -> USB 3 UVC webcam

BT.656 8-bit @ 27 MHz into the FPGA. The FPGA weaves fields in its 64 Mbit SDRAM and streams YUY2 to the FT602 (245 synchronous FIFO, 32-bit @ 100 MHz). The FT602 enumerates as UVC and tells the FPGA (its I2C slave) the selected format. The FPGA is I2C master to the TVP5150 on a separate bus.
All 66 FPGA user I/O are used: see docs/pin_assignment.csv.
Gowin settings: enable "SSPI as regular IO" and "MSPI as regular IO".

REVIEW FLAGS (not verified from the downloaded documents)
1. FT602 FPGA-side protocol (frame marking, I2C slave register map) is in FTDI AN_434: not downloaded.
2. QN88 footprint is a placeholder (ArtinChip QFN-88 10x10): check pads and EP against Gowin UG229.
3. RCA jack J1 has no footprint: pick a part (e.g. CUI RCJ-01x).
4. Crystal load caps assume CL = 18 pF: recalculate for the parts bought.
5. No anti-aliasing filter on the video input (TI recommends one).
6. USBLC6 is drawn flow-through: route D+/D- into pins 1/3 and out of 6/4.

Single sheet, fully wired. Generated by tools/design.py
Clacktronics
Sheet: /
File: 5150UVC.kicad_sch

Title: 5150UVC – composite video to USB3 UVC

Size: A2	Date: 2026-10-28	**Rev: 0.2**
KiCad E.D.A. 10.0.3		Id: 1/1